

ELITE IGCSE ACADEMY

Student Past Paper Solutions

Specimen 4WM1H Worked Solutions

Specimen | 4WM1H | Unit 1

27 questions ■ question image followed by worked solution

PREPARED BY

Dr. Eslam Ahmed

Assistant Lecturer, Cairo University Faculty of Engineering

WhatsApp: +20 112 000 9622 | eliteigcse.com

PAST PAPER QUESTION**Q#: 1****Marks: 1****TOPIC: Rounding, Estimation & Bounds**

Specimen 4WM1H - Specimen | 4WM1H | Unit 1

Answer ALL TWENTY THREE questions.**Write your answers in the spaces provided.****You must write down all stages in your working.**

- 1** The weight of a cake is 2.75 kg, correct to 2 decimal places.

Write down the lower bound of the weight of the cake.

..... kg

(Total for Question 1 is 1 mark)

EA

PAST PAPER SOLUTION**Q#: 1****Marks: 1****TOPIC: Rounding, Estimation & Bounds**

Specimen 4WM1H - Specimen | 4WM1H | Unit 1

WORKED SOLUTION*Dr. Eslam Ahmed***1. Find half the rounding interval**

Correct to 2 decimal places means the nearest 0.01 kg.

$$\frac{0.01}{2} = 0.005$$

2. Subtract from the rounded value

$$2.75 - 0.005 = 2.745$$

FINAL ANSWER

2.745 kg.

EA

PAST PAPER QUESTION**Q#: 2****Marks: 3****TOPIC: Fractions**

Specimen 4WM1H - Specimen | 4WM1H | Unit 1

2 Show that $5\frac{1}{3} - 2\frac{6}{7} = 2\frac{10}{21}$

(Total for Question 2 is 3 marks)

PAST PAPER SOLUTION**Q#: 2****Marks: 3****TOPIC: Fractions**

Specimen 4WM1H - Specimen | 4WM1H | Unit 1

WORKED SOLUTION*Dr. Eslam Ahmed***1. Convert to improper fractions**

$$5\frac{1}{3} = \frac{16}{3}, \quad 2\frac{6}{7} = \frac{20}{7}$$

2. Use a common denominator

$$\frac{16}{3} - \frac{20}{7} = \frac{112}{21} - \frac{60}{21} = \frac{52}{21}$$

3. Convert back to a mixed number

$$\frac{52}{21} = 2\frac{10}{21}$$

FINAL ANSWER

$$2\frac{10}{21}.$$

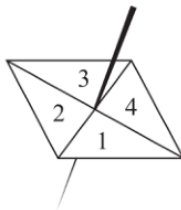
PAST PAPER QUESTION

Q#: 3 Marks: 4

TOPIC: Probability Toolkit

Specimen 4WM1H - Specimen | 4WM1H | Unit 1

3 Here is a biased 4-sided spinner.



The table gives the probabilities that, when the spinner is spun once, it will land on 1 or it will land on 3

Number	1	2	3	4
Probability	0.26		0.18	

The probability that the spinner will land on 2 is equal to the probability that the spinner will land on 4

Priya is going to spin the spinner 250 times.

Work out an estimate for the number of times the spinner will land on a 1 or a 2

(Total for Question 3 is 4 marks)

PAST PAPER SOLUTION**Q#: 3****Marks: 4****TOPIC: Probability Toolkit**

Specimen 4WM1H - Specimen | 4WM1H | Unit 1

WORKED SOLUTION*Dr. Eslam Ahmed***1. Find the remaining probability**

$$1 - 0.26 - 0.18 = 0.56$$

2. Split the equal probabilities

The probabilities for 2 and 4 are equal, so

$$P(2) = P(4) = \frac{0.56}{2} = 0.28$$

3. Find the probability of 1 or 2

$$P(1 \text{ or } 2) = 0.26 + 0.28 = 0.54$$

4. Calculate the estimate

$$0.54 \times 250 = 135$$

FINAL ANSWER

135.

PAST PAPER QUESTION**Q#: 4****Marks: 5****TOPIC: Solving Linear Equations**

Specimen 4WM1H - Specimen | 4WM1H | Unit 1

4 (a) Expand and simplify $(n - 6)(n + 4)$
(2)(b) Solve $2x - 3 = \frac{3x - 5}{4}$

Show clear algebraic working.

 $x =$
(3)

(Total for Question 4 is 5 marks)

PAST PAPER SOLUTION

Q#: 4

Marks: 5

TOPIC: **Solving Linear Equations**

Specimen 4WM1H - Specimen | 4WM1H | Unit 1

WORKED SOLUTION

Dr. Eslam Ahmed

1. Expand the brackets

$$(n - 6)(n + 4) = n^2 + 4n - 6n - 24$$

$$= n^2 - 2n - 24$$

2. Clear the denominator

$$2x - 3 = \frac{3x - 5}{4}$$

$$4(2x - 3) = 3x - 5$$

3. Solve the equation

$$8x - 12 = 3x - 5$$

$$5x = 7$$

$$x = \frac{7}{5}$$

FINAL ANSWER

$$n^2 - 2n - 24, \text{ and } x = \frac{7}{5}.$$

PAST PAPER QUESTION**Q#: 5****Marks: 3****TOPIC: Standard & Compound Units**

Specimen 4WM1H - Specimen | 4WM1H | Unit 1

5 A plane flies from New York City to Los Angeles.

The plane flies a distance of 3980 kilometres in 5 hours 24 minutes.

Work out the average speed of the plane.

Give your answer in kilometres per hour correct to the nearest whole number.

..... kilometres per hour

(Total for Question 5 is 3 marks)

PAST PAPER SOLUTION**Q#: 5****Marks: 3****TOPIC: Standard & Compound Units**

Specimen 4WM1H - Specimen | 4WM1H | Unit 1

WORKED SOLUTION*Dr. Eslam Ahmed***1. Convert the time**

$$24 \text{ minutes} = \frac{24}{60} = 0.4 \text{ hours}$$

$$5 \text{ hours } 24 \text{ minutes} = 5.4 \text{ hours}$$

2. Calculate the speed

$$\text{speed} = \frac{3980}{5.4} = 737.037 \dots$$

3. Round the answer

$$737.037 \dots \approx 737$$

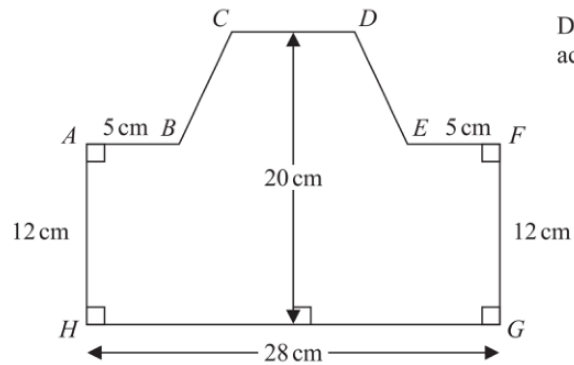
FINAL ANSWER

737 kilometres per hour.

PAST PAPER QUESTION**Q#: 6****Marks: 4****TOPIC: Area & Perimeter**

Specimen 4WM1H - Specimen | 4WM1H | Unit 1

- 6 The diagram shows an 8-sided shape $ABCDEFGH$



$$HG = 28 \text{ cm} \quad AH = FG = 12 \text{ cm} \quad AB = EF = 5 \text{ cm}$$

The height of the shape is 20 cm

CD is parallel to HG

The area of shape $ABCDEFGH$ is 434 cm^2

Find the length of CD

PAST PAPER SOLUTION**Q#: 6****Marks: 4****TOPIC: Area & Perimeter**

Specimen 4WM1H - Specimen | 4WM1H | Unit 1

WORKED SOLUTION*Dr. Eslam Ahmed***1. Find the rectangular part**

The rectangle at the bottom has width 28 cm and height 12 cm.

$$28 \times 12 = 336$$

2. Find the top area

$$434 - 336 = 98$$

3. Set up the trapezium

The top trapezium has height

$$20 - 12 = 8$$

Its lower parallel side is

$$28 - 5 - 5 = 18$$

4. Solve for the missing lengthLet $CD = x$.

$$\frac{1}{2}(x + 18) \times 8 = 98$$

$$4(x + 18) = 98$$

$$x = 6.5$$

FINAL ANSWER $CD = 6.5$ cm.

PAST PAPER QUESTION**Q#: 7****Marks: 4****TOPIC: Algebraic Roots & Indices**

Specimen 4WM1H - Specimen | 4WM1H | Unit 1

7 (a) Simplify $8 \times (4t)^0$
(1)

$$x^6 \div x^{-5} = x^p$$

(b) Find the value of p $p =$
(1)(c) Simplify fully $(2k^2m^4)^3$
(2)

PAST PAPER SOLUTION**Q#: 7****Marks: 4****TOPIC: Algebraic Roots & Indices**

Specimen 4WM1H - Specimen | 4WM1H | Unit 1

WORKED SOLUTION*Dr. Eslam Ahmed***1. Use the zero power**

$$(4t)^0 = 1$$

$$8(4t)^0 = 8$$

2. Subtract the indices

$$x^6 \div x^{-5} = x^{6-(-5)} = x^{11}$$

So $p = 11$.**3. Raise each factor to the power**

$$(2k^2m^4)^3 = 2^3k^{2 \times 3}m^{4 \times 3}$$

$$= 8k^6m^{12}$$

FINAL ANSWER8, $p = 11$, and $8k^6m^{12}$.

PAST PAPER QUESTION**Q#: 8****Marks: 3****TOPIC: Standard & Compound Units**

Specimen 4WM1H - Specimen | 4WM1H | Unit 1

(Total for Question 7 is 4 marks)

8 Change a speed of 81 kilometres per hour to a speed in metres per second.

..... metres per second

(Total for Question 8 is 3 marks)

PAST PAPER SOLUTION**Q#: 8****Marks: 3****TOPIC: Standard & Compound Units**

Specimen 4WM1H - Specimen | 4WM1H | Unit 1

WORKED SOLUTION*Dr. Eslam Ahmed***1. Convert kilometres to metres**

$$81 \text{ km} = 81000 \text{ m}$$

2. Convert hours to seconds

$$1 \text{ hour} = 3600 \text{ seconds}$$

3. Divide to find metres per second

$$\frac{81000}{3600} = 22.5$$

FINAL ANSWER

22.5 metres per second.

PAST PAPER QUESTION**Q#: 9** **Marks: 7****TOPIC: Linear Graphs ($y = mx + c$)**

Specimen 4WM1H - Specimen | 4WM1H | Unit 1

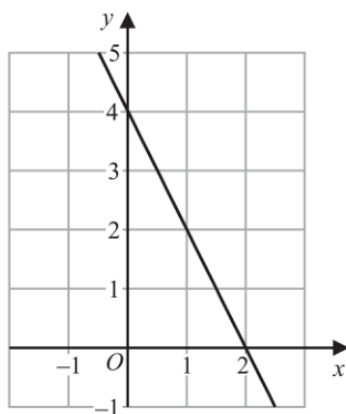
- 9 (a) Simplify
- $3a^4b^5 \times 4a^7b^2$

.....
(2)

- (b) Factorise fully
- $14x^2y^4 + 21x^3y^2$

.....
(2)

The diagram shows a straight line drawn on a grid.



- (c) Find an equation of the line.

.....
(2)A different straight line has equation $y = 3x - 5$

- (d) Write down the coordinates of the point at which the line crosses the y-axis.

(.....,)
(1)**(Total for Question 9 is 7 marks)**

PAST PAPER SOLUTION**Q#: 9****Marks: 7****TOPIC: Linear Graphs ($y = mx + c$)**

Specimen 4WM1H - Specimen | 4WM1H | Unit 1

WORKED SOLUTION*Dr. Eslam Ahmed***1. Simplify the product**

$$3a^4b^5 \times 4a^7b^2 = 12a^{11}b^7$$

2. Factorise fully

$$14x^2y^4 + 21x^3y^2 = 7x^2y^2(2y^2 + 3x)$$

3. Find the gradient

The line passes through (0, 4) and (2, 0).

$$m = \frac{0 - 4}{2 - 0} = -2$$

4. Write the equationThe y -intercept is 4, so

$$y = -2x + 4$$

5. Read the y -axis crossingFor $y = 3x - 5$, put $x = 0$:

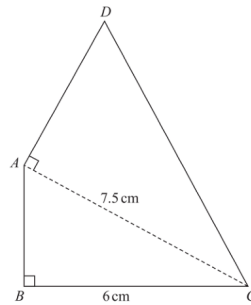
$$y = -5$$

So the crossing point is (0, -5).

FINAL ANSWER $12a^{11}b^7$, $7x^2y^2(2y^2 + 3x)$, $y = -2x + 4$, and (0, -5).

PAST PAPER QUESTION**Q#: 10****Marks: 6****TOPIC: Right-Angled Triangles - Pythagoras & Trigonometry**

Specimen 4WM1H - Specimen | 4WM1H | Unit 1

10 The diagram shows a quadrilateral $ABCD$ Diagram **NOT**
accurately drawnIn the diagram, ABC and DAC are right-angled triangles.

$$BC = 6 \text{ cm} \quad AC = 7.5 \text{ cm}$$

The area of quadrilateral $ABCD$ is 31.5 cm^2 (a) Work out the length of AB cm
(2)(b) Work out the length of AD cm
(4)

(Total for Question 10 is 6 marks)

PAST PAPER SOLUTION**Q#: 10****Marks: 6****TOPIC: Right-Angled Triangles - Pythagoras & Trigonometry**

Specimen 4WM1H - Specimen | 4WM1H | Unit 1

WORKED SOLUTION*Dr. Eslam Ahmed***1. Use Pythagoras for AB**In triangle ABC ,

$$AB^2 + 6^2 = 7.5^2$$

$$AB^2 = 56.25 - 36 = 20.25$$

$$AB = 4.5$$

2. Find the area of triangle ABC

$$\frac{1}{2} \times 6 \times 4.5 = 13.5$$

3. Find the area of triangle DAC

$$31.5 - 13.5 = 18$$

4. Use the triangle area formulaIn triangle DAC , AD and AC are perpendicular.

$$\frac{1}{2} \times AD \times 7.5 = 18$$

$$AD = \frac{36}{7.5} = 4.8$$

FINAL ANSWER $AB = 4.5$ cm, $AD = 4.8$ cm.

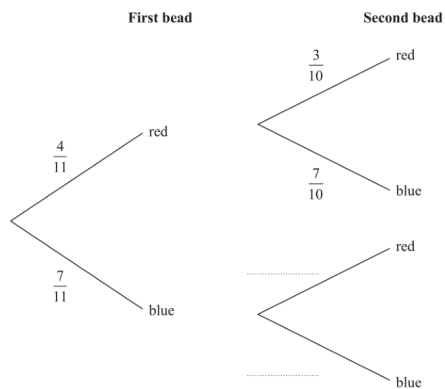
PAST PAPER QUESTION**Q#: 11****Marks: 6****TOPIC: Probability Diagrams - Venn & Tree Diagrams**

Specimen 4WM1H - Specimen | 4WM1H | Unit 1

- 11 Tess has a bag containing 11 beads.
4 of the beads are red.
7 of the beads are blue.

Tess takes at random a bead from the bag and keeps it.
Tess takes at random another bead from the bag and keeps it.

- (a) Complete the probability tree diagram.



(1)

- (b) Work out the probability that both of the beads are red.

(2)

- (c) Work out the probability that the two beads are of different colours.

(3)

(Total for Question 11 is 6 marks)

PAST PAPER SOLUTION**Q#: 11****Marks: 6****TOPIC: Probability Diagrams - Venn & Tree Diagrams**

Specimen 4WM1H - Specimen | 4WM1H | Unit 1

WORKED SOLUTION*Dr. Eslam Ahmed***1. Complete the blue-first branch**

After a blue bead is taken first, there are 4 red and 6 blue beads left out of 10.

$$P(\text{red second after blue}) = \frac{4}{10}, \quad P(\text{blue second after blue}) = \frac{6}{10}$$

2. Find both red

$$P(RR) = \frac{4}{11} \times \frac{3}{10} = \frac{12}{110} = \frac{6}{55}$$

3. Find different colours

$$P(RB) = \frac{4}{11} \times \frac{7}{10}$$

$$P(BR) = \frac{7}{11} \times \frac{4}{10}$$

$$P(\text{different}) = \frac{28}{110} + \frac{28}{110} = \frac{28}{55}$$

FINAL ANSWER

Missing probabilities $\frac{4}{10}$ and $\frac{6}{10}$; $P(RR) = \frac{6}{55}$; $P(\text{different}) = \frac{28}{55}$.

PAST PAPER QUESTION**Q#: 12****Marks: 5****TOPIC: Algebraic Fractions**

Specimen 4WM1H - Specimen | 4WM1H | Unit 1

12 (a) Factorise $9x^2 - 4y^2$

(2)**(b)** Express $\frac{7}{8} - \frac{x+3}{4x}$ as a single fraction in its simplest form.

(3)

(Total for Question 12 is 5 marks)

PAST PAPER SOLUTION**Q#: 12****Marks: 5****TOPIC: Algebraic Fractions**

Specimen 4WM1H - Specimen | 4WM1H | Unit 1

WORKED SOLUTION*Dr. Eslam Ahmed***1. Use difference of two squares**

$$\begin{aligned}
 9x^2 - 4y^2 &= (3x)^2 - (2y)^2 \\
 &= (3x - 2y)(3x + 2y)
 \end{aligned}$$

2. Use a common denominator

$$\frac{7}{8} - \frac{x+3}{4x} = \frac{7x}{8x} - \frac{2(x+3)}{8x}$$

3. Simplify the numerator

$$\frac{7x - 2x - 6}{8x} = \frac{5x - 6}{8x}$$

FINAL ANSWER $(3x - 2y)(3x + 2y)$, and $\frac{5x-6}{8x}$.

PAST PAPER QUESTION**Q#: 13****Marks: 6****TOPIC: Algebraic Roots & Indices**

Specimen 4WM1H - Specimen | 4WM1H | Unit 1

13 (a) Expand and simplify $(3x - 1)(x + 2)(3x + 1)$ **(b)** Simplify fully $\left(\frac{2x^5}{8xy^2}\right)^{-2}$
(3).....
(3)**(Total for Question 13 is 6 marks)**

PAST PAPER SOLUTION**Q#: 13****Marks: 6****TOPIC: Algebraic Roots & Indices**

Specimen 4WM1H - Specimen | 4WM1H | Unit 1

WORKED SOLUTION*Dr. Eslam Ahmed***1. Multiply the conjugate factors**

$$(3x - 1)(3x + 1) = 9x^2 - 1$$

2. Expand the remaining bracket

$$(9x^2 - 1)(x + 2) = 9x^3 + 18x^2 - x - 2$$

3. Simplify inside the bracket

$$\frac{2x^5}{8xy^2} = \frac{x^4}{4y^2}$$

4. Apply the negative power

$$\left(\frac{x^4}{4y^2}\right)^{-2} = \left(\frac{4y^2}{x^4}\right)^2 = \frac{16y^4}{x^8}$$

FINAL ANSWER $9x^3 + 18x^2 - x - 2$, and $\frac{16y^4}{x^8}$.

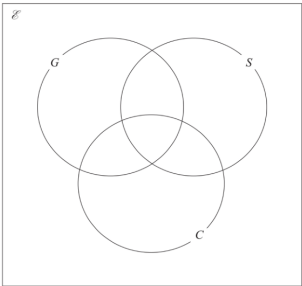
PAST PAPER QUESTION

Q#: 14 Marks: 8

TOPIC: Set Notation & Venn Diagrams

Specimen 4WM1H - Specimen | 4WM1H | Unit 1

- 14 100 farmers are asked if they have goats (G), sheep (S) or chickens (C) on their farms.
- Of these farmers
- 31 have sheep
 - 53 have chickens
 - 6 have goats, sheep and chickens
 - 11 have sheep and goats
 - 17 have sheep and chickens
 - 18 have goats and chickens
 - 20 do not have any goats, sheep or chickens
- (a) Using this information, complete the Venn diagram to show the number of farmers in each appropriate subset.



(3)

- (b) Find
- (i) $n(G)$

(1)

- (ii) $n([G \cup S]')$

(1)

- (iii) $n(G' \cap C)$

(1)

- One of the farmers who has chickens is chosen at random.
- (c) Find the probability that this farmer also has goats.

(2)

(Total for Question 14 is 8 marks)

PAST PAPER SOLUTION**Q#: 14****Marks: 8****TOPIC: Set Notation & Venn Diagrams**

Specimen 4WM1H - Specimen | 4WM1H | Unit 1

WORKED SOLUTION*Dr. Eslam Ahmed***1. Place the centre value**

The number with goats, sheep and chickens is 6.

2. Fill the pair-only regions

$$G \cap S \text{ only} = 11 - 6 = 5$$

$$S \cap C \text{ only} = 17 - 6 = 11$$

$$G \cap C \text{ only} = 18 - 6 = 12$$

3. Fill the single-set regions

$$S \text{ only} = 31 - 5 - 11 - 6 = 9$$

$$C \text{ only} = 53 - 11 - 12 - 6 = 24$$

Since 20 have none, the total inside the circles is 80.

$$G \text{ only} = 80 - (5 + 11 + 12 + 6 + 9 + 24) = 13$$

4. Read the set values

$$n(G) = 13 + 5 + 12 + 6 = 36$$

$$n((G \cup S)') = 24 + 20 = 44$$

$$n(G' \cap C) = 24 + 11 = 35$$

5. Find the conditional probability

Out of the 53 farmers with chickens, 18 also have goats.

$$P(G | C) = \frac{18}{53}$$

FINAL ANSWER

Venn regions: G only 13, S only 9, C only 24, $G \cap S$ only 5, $S \cap C$ only 11, $G \cap C$ only 12, all three 6, none 20. Also 36, 44, 35, and $\frac{18}{53}$.

PAST PAPER QUESTION**Q#: 15****Marks: 4****TOPIC: Rounding, Estimation & Bounds**

Specimen 4WM1H - Specimen | 4WM1H | Unit 1

15 Martin and Lucia went on two different car journeys.

For Martin's journey

distance = 80 km correct to the nearest 5 km

time = 2.7 hours correct to 1 decimal place

For Lucia's journey

distance = 33 km correct to 2 significant figures

time = 1 hour correct to the nearest 0.1 hour

Martin says,

"My average speed could have been greater than Lucia's average speed."

By considering bounds, show that Martin is correct.

Show your working clearly.

(Total for Question 15 is 4 marks)

PAST PAPER SOLUTION**Q#: 15****Marks: 4****TOPIC: Rounding, Estimation & Bounds**

Specimen 4WM1H - Specimen | 4WM1H | Unit 1

WORKED SOLUTION*Dr. Eslam Ahmed***1. Use Martin's greatest possible speed**

Martin's greatest possible distance is 82.5 km and his least possible time is 2.65 hours.

$$\frac{82.5}{2.65} = 31.132 \dots$$

2. Use Lucia's least possible speed

Lucia's least possible distance is 32.5 km and her greatest possible time is 1.05 hours.

$$\frac{32.5}{1.05} = 30.952 \dots$$

3. Compare the possible speeds

$$31.132 \dots > 30.952 \dots$$

So Martin's speed could have been greater than Lucia's speed.

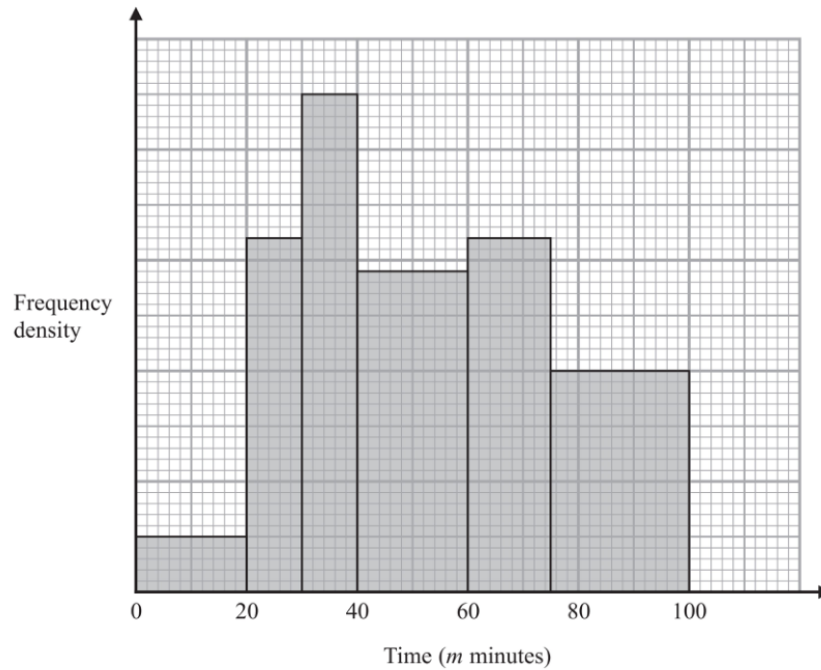
FINAL ANSWER

Martin could have been faster because $31.132 \dots > 30.952 \dots$ km/h.

PAST PAPER QUESTION**Q#: 16****Marks: 3****TOPIC: Histograms**

Specimen 4WM1H - Specimen | 4WM1H | Unit 1

- 16** The histogram shows information about the total time, m minutes, taken by each child in a school to walk to school every day for one week.



There are no children for whom $m > 100$

There are 10 children for whom $m \leq 20$

Work out an estimate for the number of children for whom $50 < m \leq 80$

(Total for Question 16 is 3 marks)

PAST PAPER SOLUTION**Q#: 16****Marks: 3****TOPIC: Histograms**

Specimen 4WM1H - Specimen | 4WM1H | Unit 1

WORKED SOLUTION*Dr. Eslam Ahmed***1. Set the frequency-density scale**For $0 < m \leq 20$, the frequency is 10, so

$$\text{frequency density} = \frac{10}{20} = 0.5$$

2. Read the needed bar heights

Using the same scale from the histogram:

$$50 < m \leq 60 : 10 \times 2.9 = 29$$

$$60 < m \leq 75 : 15 \times 3.2 = 48$$

$$75 < m \leq 80 : 5 \times 2 = 10$$

3. Add the estimates

$$19 + 36 + 32 = 87$$

FINAL ANSWER

87.

PAST PAPER QUESTION**Q#: 17****Marks: 4****TOPIC: Surds**

Specimen 4WM1H - Specimen | 4WM1H | Unit 1

17 Express $\frac{3 + \sqrt{8}}{(\sqrt{2} - 1)^2}$ in the form $p + \sqrt{q}$ where p and q are integers.

Show clearly each stage of your working.

.....
(Total for Question 17 is 4 marks)

PAST PAPER SOLUTION**Q#: 17****Marks: 4****TOPIC: Surds**

Specimen 4WM1H - Specimen | 4WM1H | Unit 1

WORKED SOLUTION*Dr. Eslam Ahmed***1. Simplify the surds**

$$\sqrt{8} = 2\sqrt{2}$$

$$(\sqrt{2} - 1)^2 = 2 - 2\sqrt{2} + 1 = 3 - 2\sqrt{2}$$

2. Use the conjugate

$$\frac{3 + 2\sqrt{2}}{3 - 2\sqrt{2}} \times \frac{3 + 2\sqrt{2}}{3 + 2\sqrt{2}}$$

The denominator is

$$9 - (2\sqrt{2})^2 = 9 - 8 = 1$$

3. Expand the numerator

$$(3 + 2\sqrt{2})^2 = 9 + 12\sqrt{2} + 8 = 17 + 12\sqrt{2}$$

4. Write in the requested form

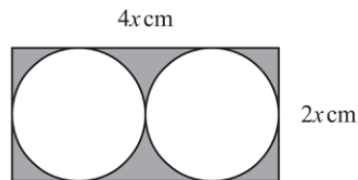
$$12\sqrt{2} = \sqrt{144 \times 2} = \sqrt{288}$$

FINAL ANSWER

$$17 + \sqrt{288}.$$

PAST PAPER QUESTION**Q#: 18****Marks: 4****TOPIC: Circles, Arcs & Sectors**

Specimen 4WM1H - Specimen | 4WM1H | Unit 1

18 The diagram shows two identical circles drawn inside a rectangle.Diagram **NOT**
accurately drawn

The length of the rectangle is $4x$ cm and the width of the rectangle is $2x$ cm
 Each circle has a radius of x cm

The circles touch each other and each circle touches three sides of the rectangle.

The region inside the rectangle that is outside the circles, shown shaded in the diagram, has a total area of 20 cm^2

Work out the perimeter of the rectangle.
 Give your answer correct to 3 significant figures.

.....cm

PAST PAPER SOLUTION**Q#: 18****Marks: 4****TOPIC: Circles, Arcs & Sectors**

Specimen 4WM1H - Specimen | 4WM1H | Unit 1

WORKED SOLUTION*Dr. Eslam Ahmed***1. Write the shaded area**

Rectangle area:

$$4x \times 2x = 8x^2$$

Area of the two circles:

$$2\pi x^2$$

So

$$8x^2 - 2\pi x^2 = 20$$

2. Solve for x

$$x^2(8 - 2\pi) = 20$$

$$x = \sqrt{\frac{20}{8 - 2\pi}} = 3.413\dots$$

3. Find the perimeter

$$P = 2(4x + 2x) = 12x$$

$$12(3.413\dots) = 40.96\dots$$

4. Round to 3 significant figures

$$40.96\dots \approx 41.0$$

FINAL ANSWER

41.0 cm.

PAST PAPER QUESTION**Q#: 19****Marks: 4****TOPIC: 3D Pythagoras & Trigonometry**

Specimen 4WM1H - Specimen | 4WM1H | Unit 1

19 The diagram shows a triangular prism, $ABCDEF$, with a rectangular base $ABCD$

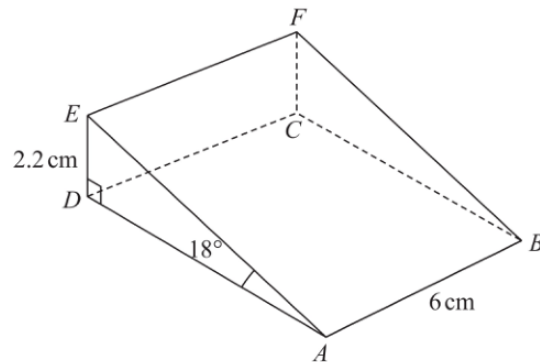


Diagram **NOT**
accurately drawn

$AB = 6 \text{ cm}$ $DE = 2.2 \text{ cm}$ $\text{angle } DAE = 18^\circ$ $\text{angle } ADE = 90^\circ$

Work out the angle that BE makes with the plane $ABCD$
Give your answer correct to one decimal place.

◦

PAST PAPER SOLUTION**Q#: 19****Marks: 4****TOPIC: 3D Pythagoras & Trigonometry**

Specimen 4WM1H - Specimen | 4WM1H | Unit 1

WORKED SOLUTION*Dr. Eslam Ahmed***1. Find AD**In triangle ADE ,

$$\tan 18^\circ = \frac{DE}{AD}$$

$$AD = \frac{2.2}{\tan 18^\circ} = 6.77 \dots$$

2. Find DB in the baseThe base $ABCD$ is a rectangle, so

$$DB = \sqrt{6^2 + 6.77 \dots^2} = 9.04 \dots$$

3. Use the projection triangleThe projection of BE on the plane is BD . Let the required angle be θ .

$$\tan \theta = \frac{DE}{DB} = \frac{2.2}{9.04 \dots}$$

4. Calculate the angle

$$\theta = 13.7^\circ$$

FINAL ANSWER

13.7°.

PAST PAPER QUESTION**Q#: 20****Marks: 4****TOPIC: Completing the Square**

Specimen 4WM1H - Specimen | 4WM1H | Unit 1

20 Find the values of a , b and c so that

$$7 + 12x - 2x^2$$

is written as $a - b(x - c)^2$

$$a = \dots\dots\dots$$

$$b = \dots\dots\dots$$

$$c = \dots\dots\dots$$

(Total for Question 20 is 4 marks)

PAST PAPER SOLUTION**Q#: 20****Marks: 4****TOPIC: Completing the Square**

Specimen 4WM1H - Specimen | 4WM1H | Unit 1

WORKED SOLUTION*Dr. Eslam Ahmed***1. Factor out -2 from the quadratic terms**

$$7 + 12x - 2x^2 = -2(x^2 - 6x) + 7$$

2. Complete the square

$$x^2 - 6x = (x - 3)^2 - 9$$

3. Simplify the expression

$$\begin{aligned} -2((x - 3)^2 - 9) + 7 &= -2(x - 3)^2 + 18 + 7 \\ &= 25 - 2(x - 3)^2 \end{aligned}$$

4. Read the values

$$a = 25, \quad b = 2, \quad c = 3$$

FINAL ANSWER

$$a = 25, \quad b = 2, \quad c = 3.$$

PAST PAPER QUESTION**Q#: 20****Marks: 4****TOPIC: Completing the Square**

Specimen 4WM1H - Specimen | 4WM1H | Unit 1

20 Find the values of a , b and c so that

$$7 + 12x - 2x^2$$

is written as $a - b(x - c)^2$

$$a = \dots\dots\dots$$

$$b = \dots\dots\dots$$

$$c = \dots\dots\dots$$

(Total for Question 20 is 4 marks)

PAST PAPER SOLUTION**Q#: 20****Marks: 4****TOPIC: Completing the Square**

Specimen 4WM1H - Specimen | 4WM1H | Unit 1

WORKED SOLUTION*Dr. Eslam Ahmed***1. Factor out -2 from the quadratic terms**

$$7 + 12x - 2x^2 = -2(x^2 - 6x) + 7$$

2. Complete the square

$$x^2 - 6x = (x - 3)^2 - 9$$

3. Simplify the expression

$$\begin{aligned} -2((x - 3)^2 - 9) + 7 &= -2(x - 3)^2 + 18 + 7 \\ &= 25 - 2(x - 3)^2 \end{aligned}$$

4. Read the values

$$a = 25, \quad b = 2, \quad c = 3$$

FINAL ANSWER

$$a = 25, \quad b = 2, \quad c = 3.$$

PAST PAPER QUESTION**Q#: 21****Marks: 3****TOPIC: Algebraic Fractions**

Specimen 4WM1H - Specimen | 4WM1H | Unit 1

21 Express $\left(\frac{20}{x^2 - 36} - \frac{2}{x - 6}\right) \times \frac{1}{4 - x}$ as a single fraction in its simplest form.

Show clear algebraic working.

PAST PAPER SOLUTION**Q#: 21****Marks: 3****TOPIC: Algebraic Fractions**

Specimen 4WM1H - Specimen | 4WM1H | Unit 1

WORKED SOLUTION*Dr. Eslam Ahmed***1. Factorise the denominator**

$$x^2 - 36 = (x - 6)(x + 6)$$

2. Simplify the bracket

$$\begin{aligned} \frac{20}{(x-6)(x+6)} - \frac{2}{x-6} &= \frac{20 - 2(x+6)}{(x-6)(x+6)} \\ &= \frac{8 - 2x}{(x-6)(x+6)} = \frac{-2(x-4)}{(x-6)(x+6)} \end{aligned}$$

3. Cancel the common factorSince $4 - x = -(x - 4)$,

$$\frac{-2(x-4)}{(x-6)(x+6)} \times \frac{1}{4-x} = \frac{2}{(x-6)(x+6)}$$

FINAL ANSWER

$$\frac{2}{x^2 - 36}$$

PAST PAPER QUESTION**Q#: 21****Marks: 3****TOPIC: Algebraic Fractions**

Specimen 4WM1H - Specimen | 4WM1H | Unit 1

21 Express $\left(\frac{20}{x^2 - 36} - \frac{2}{x - 6}\right) \times \frac{1}{4 - x}$ as a single fraction in its simplest form.

Show clear algebraic working.

PAST PAPER SOLUTION**Q#: 21****Marks: 3****TOPIC: Algebraic Fractions**

Specimen 4WM1H - Specimen | 4WM1H | Unit 1

WORKED SOLUTION*Dr. Eslam Ahmed***1. Factorise the denominator**

$$x^2 - 36 = (x - 6)(x + 6)$$

2. Simplify the bracket

$$\begin{aligned} \frac{20}{(x-6)(x+6)} - \frac{2}{x-6} &= \frac{20 - 2(x+6)}{(x-6)(x+6)} \\ &= \frac{8 - 2x}{(x-6)(x+6)} = \frac{-2(x-4)}{(x-6)(x+6)} \end{aligned}$$

3. Cancel the common factorSince $4 - x = -(x - 4)$,

$$\frac{-2(x-4)}{(x-6)(x+6)} \times \frac{1}{4-x} = \frac{2}{(x-6)(x+6)}$$

FINAL ANSWER

$$\frac{2}{x^2 - 36}$$

PAST PAPER QUESTION**Q#: 22****Marks: 3****TOPIC: Algebraic Roots & Indices**

Specimen 4WM1H - Specimen | 4WM1H | Unit 1

22 $\frac{100 \sqrt{2n+8}}{6 \times 9^{2n+8}} = 3^x$

Express x in terms of n

Show your working clearly and simplify your expression.

 $x = \dots\dots\dots$

PAST PAPER SOLUTION**Q#: 22****Marks: 3****TOPIC: Algebraic Roots & Indices**

Specimen 4WM1H - Specimen | 4WM1H | Unit 1

WORKED SOLUTION*Dr. Eslam Ahmed***1. Write everything as powers of 3**

$$\sqrt{27} = 3^{3/2}, \quad 9 = 3^2, \quad \frac{18}{6} = 3$$

2. Simplify the powers

$$\begin{aligned} \frac{18(\sqrt{27})^{4n+6}}{6(9)^{2n+8}} &= 3 \times \frac{(3^{3/2})^{4n+6}}{(3^2)^{2n+8}} \\ &= 3^1 \times \frac{3^{6n+9}}{3^{4n+16}} \end{aligned}$$

3. Subtract the indices

$$3^{1+6n+9-(4n+16)} = 3^{2n-6}$$

So

$$x = 2n - 6$$

FINAL ANSWER

$$x = 2n - 6.$$

PAST PAPER QUESTION**Q#: 22****Marks: 3****TOPIC: Algebraic Roots & Indices**

Specimen 4WM1H - Specimen | 4WM1H | Unit 1

22 $\frac{100 \sqrt{2n+8}}{6 \times 9^{2n+8}} = 3^x$

Express x in terms of n

Show your working clearly and simplify your expression.

 $x = \dots\dots\dots$

PAST PAPER SOLUTION**Q#: 22****Marks: 3****TOPIC: Algebraic Roots & Indices**

Specimen 4WM1H - Specimen | 4WM1H | Unit 1

WORKED SOLUTION*Dr. Eslam Ahmed***1. Write everything as powers of 3**

$$\sqrt{27} = 3^{3/2}, \quad 9 = 3^2, \quad \frac{18}{6} = 3$$

2. Simplify the powers

$$\begin{aligned} \frac{18(\sqrt{27})^{4n+6}}{6(9)^{2n+8}} &= 3 \times \frac{(3^{3/2})^{4n+6}}{(3^2)^{2n+8}} \\ &= 3^1 \times \frac{3^{6n+9}}{3^{4n+16}} \end{aligned}$$

3. Subtract the indices

$$3^{1+6n+9-(4n+16)} = 3^{2n-6}$$

So

$$x = 2n - 6$$

FINAL ANSWER

$$x = 2n - 6.$$

PAST PAPER QUESTION**Q#: 23****Marks: 6****TOPIC: Coordinate Geometry**

Specimen 4WM1H - Specimen | 4WM1H | Unit 1

23 $ABCD$ is a kite.

$$AB = AD \text{ and } CB = CD$$

The point B has coordinates $(k, 1)$ where k is a negative constant.The point D has coordinates $(8, 7)$ The straight line L passes through the points B and D and has a gradient of $\frac{3}{5}$ Find an equation of AC Give your answer in the form $px + qy = r$ where p , q and r are integers.

Show your working clearly.

(Total for Question 23 is 6 marks)

PAST PAPER SOLUTION**Q#: 23****Marks: 6****TOPIC: Coordinate Geometry**

Specimen 4WM1H - Specimen | 4WM1H | Unit 1

WORKED SOLUTION*Dr. Eslam Ahmed***1. Find k from the gradient**The gradient of BD is

$$\frac{7-1}{8-k} = \frac{3}{5}$$

$$\frac{6}{8-k} = \frac{3}{5}$$

$$30 = 24 - 3k$$

$$k = -2$$

2. Find the midpoint of BDSo $B = (-2, 1)$ and $D = (8, 7)$.

$$\text{midpoint of } BD = \left(\frac{-2+8}{2}, \frac{1+7}{2} \right) = (3, 4)$$

3. Use the perpendicular gradientIn this kite, AC is the perpendicular bisector of BD .The gradient of BD is $\frac{3}{5}$, so the gradient of AC is

$$-\frac{5}{3}$$

4. Form the equationUsing point $(3, 4)$:

$$y - 4 = -\frac{5}{3}(x - 3)$$

$$3y - 12 = -5x + 15$$

$$5x + 3y = 27$$

FINAL ANSWER

$$5x + 3y = 27.$$

PAST PAPER QUESTION**Q#: 23****Marks: 6****TOPIC: Coordinate Geometry**

Specimen 4WM1H - Specimen | 4WM1H | Unit 1

23 $ABCD$ is a kite.

$$AB = AD \text{ and } CB = CD$$

The point B has coordinates $(k, 1)$ where k is a negative constant.The point D has coordinates $(8, 7)$ The straight line L passes through the points B and D and has a gradient of $\frac{3}{5}$ Find an equation of AC Give your answer in the form $px + qy = r$ where p , q and r are integers.

Show your working clearly.

(Total for Question 23 is 6 marks)

PAST PAPER SOLUTION**Q#: 23****Marks: 6****TOPIC: Coordinate Geometry**

Specimen 4WM1H - Specimen | 4WM1H | Unit 1

WORKED SOLUTION*Dr. Eslam Ahmed***1. Find k from the gradient**The gradient of BD is

$$\frac{7-1}{8-k} = \frac{3}{5}$$

$$\frac{6}{8-k} = \frac{3}{5}$$

$$30 = 24 - 3k$$

$$k = -2$$

2. Find the midpoint of BDSo $B = (-2, 1)$ and $D = (8, 7)$.

$$\text{midpoint of } BD = \left(\frac{-2+8}{2}, \frac{1+7}{2} \right) = (3, 4)$$

3. Use the perpendicular gradientIn this kite, AC is the perpendicular bisector of BD .The gradient of BD is $\frac{3}{5}$, so the gradient of AC is

$$-\frac{5}{3}$$

4. Form the equationUsing point $(3, 4)$:

$$y - 4 = -\frac{5}{3}(x - 3)$$

$$3y - 12 = -5x + 15$$

$$5x + 3y = 27$$

FINAL ANSWER

$$5x + 3y = 27.$$